Name: _____

TOPIC TEST 1

Chapter 4: Functions and graphs

- Time allowed: 45 minutes
- CAS calculators may be used.
- Part A: 5 multiple-choice questions (5 marks)
- Part B: 7 short answer questions (49 marks)
- Part C: 2 analysis questions (16 marks)

Part A

5 multiple-choice questions
1 mark each: 5 marks
Circle the correct answer.

1 Which of the following is NOT a function?

A $y = \dfrac{1}{x+3}$

B $x = y^2 - 7y + 1$

C $x^2 - y + 1 = 0$

D $3x - y - 1 = 0$

E $y = x^4 + 8$

[1 mark]

2 Which function is discontinuous?

A $y = \dfrac{1}{x^2+5}$

B $y = x^2 - 5$

C $f(x) = \dfrac{1}{x^2-5}$

D $f(x) = \sqrt{x^2+5}$

E $y = \begin{cases} x^2 - 5 & \text{if } x \geq 0 \\ -x - 5 & \text{if } x < 0 \end{cases}$

[1 mark]

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3 Which is the equation of the function 2 units to the right of $y = x^3$ and 3 units up?

A $y = (x - 2)^3 - 3$

B $y = (x + 2)^3 - 3$

C $y = (x - 2)^3 + 3$

D $y = 2x^2 - 3$

E $y = (x + 2)^3 + 3$

[1 mark]

4 Find the gradient of the line that is perpendicular to the straight line with equation $2x + 6y - 1 = 0$

A -3

B $\frac{1}{3}$

C $-\frac{1}{2}$

D $-\frac{1}{3}$

E 3

[1 mark]

5 Find the x-intercepts of $p(x) = 2x^3 + 9x^2 - 5x$

A $-5, 0, \frac{1}{2}$

B $-\frac{1}{2}, 0, 5$

C $0, 1, 5$

D $0, 2, 5$

E $-5, 0, 2$

[1 mark]

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Part B

7 short answer questions
49 marks
Show your working where appropriate.

1 Sketch the following graphs and state their domain and range.

a $x^2 + y^2 = 9$

b $y^2 = -2x$

c $y = -\dfrac{1}{x+2}$

d $y = (x-2)(x+1)(x+2)$

e $f(x) = (x-4)^2$

[15 marks]

2 Find the equation of each of the following.

a The line with x-intercept 4 and gradient 3.

b The quadratic function with turning point $(-2, -1)$ and y-intercept 11.

c The circle with centre $(-3, 2)$ and radius 2.

[9 marks]

3 For the function $f(x) = x^3 - 4x$:

a prove that the function is odd

b evaluate $f(-1)$

c find the x- and y-intercepts of the function

d sketch the function.

[7 marks]

4 For the given points $A(-2, 3)$ and $B(-1, -4)$:

a find the midpoint M of A and B

b find the distance between A and B

c find the gradient of line AB.

[6 marks]

5 For the function $y = x^2 - 4x - 5$:

a find the discriminant and state the number and type of its zeros

b find the equation of its axis of symmetry

c find the coordinates of its turning point and state whether it is a maximum or minimum.

[6 marks]

6 Find the radius and the coordinates of the centre of the circle $x^2 + 4x + y^2 - 6y - 12 = 0$.

[4 marks]

7 Prove that the line $y = 3x - 1$ and $6x - 2y - 5 = 0$ are parallel.

[2 marks]

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Part C

2 analysis questions
16 marks
Show your working where appropriate.

1 For the function $y = ax^2 + bx + c$:

a state the x-intercepts

b show that the equation of the axis of symmetry is $x = -\frac{b}{2a}$.

[4 marks]

2 **a** Show that the line $3x - 4y + 5 = 0$ passes through the circle $x^2 + y^2 = 25$ at two points.

b Find these points of intersection correct to one decimal place (you may use a CAS calculator).

[12 marks]
Total marks: 60

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Answers

Part A **[1 mark each]**

1 For **A**, **C**, **D**, and **E**, there is only one y-value for every x value. In **B** an x-value can take on two different y-values, e.g. when $x = 1$, $y = 0$ or 7.

$\therefore$ **B** **[1 mark]**

2 The data is continuous for all functions except **C** where $x^2 - 5 \neq 0$. There are asymptotes at $x = \pm\sqrt{5}$ and the function is discontinuous at these points.

$\therefore$ **C** **[1 mark]**

3 $y = (x - 2)^3$ is a horizontal translation of $y = x^3$ by 2 units to the right.

$y = x^3 + 3$ is a vertical translation of $y = x^3$ by 3 units upwards.

So $y = (x - 2)^3 + 3$ moves the function 2 units to the right and 3 units up.

$\therefore$ **C** **[1 mark]**

4 Writing $2x + 6y - 1 = 0$ in gradient−intercept form:

$6y = -2x + 1$

$y = -\frac{1}{3}x + \frac{1}{6}$

Gradient $m_1 = -\frac{1}{3}$

For perpendicular lines: $m_1 m_2 = -1$

So $m_2 = 3$

$\therefore$ **E** **[1 mark]**

5 For x-intercepts, $f(x) = 0$ $(y = 0)$

$2x^3 + 9x^2 - 5x = 0$

$x(2x^2 + 9x - 5) = 0$

$x(2x - 1)(x + 5) = 0$

$x = 0, 2x - 1 = 0, x + 5 = 0$

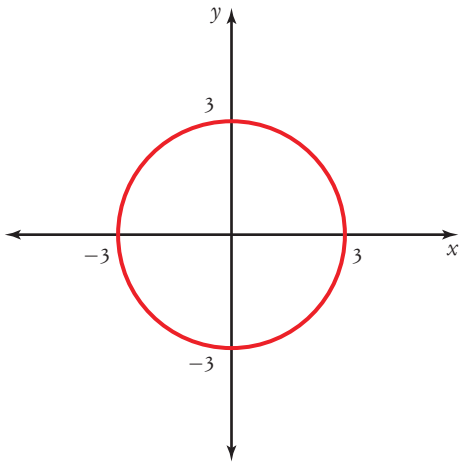
$x = 0, \frac{1}{2}, -5$

$\therefore$ **A** **[1 mark]**

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Part B

1 a This is a circle with centre (0, 0) and radius $\sqrt{9}$ or 3 units.

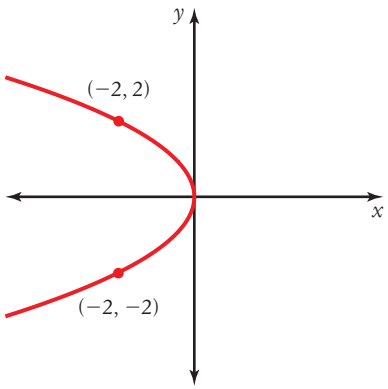


[1 mark]

Domain is $\{x: -3 \leq x \leq 3\}$; range is $\{y: -3 \leq y \leq 3\}$. **[1 mark]**

b This is a parabola with a horizontal axis of symmetry.

$y^2 = -2x$ gives two functions $y = \pm\sqrt{-2x}$, where $x \leq 0$.

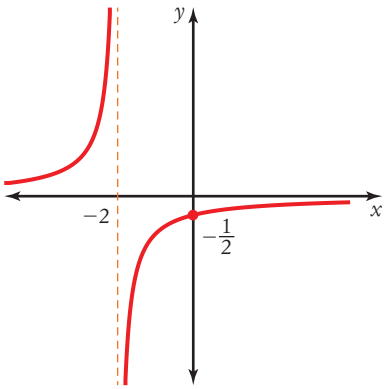


[2 marks]

Domain is $\{x:$ all real $x \leq 0\}$; range is $\{y:$ all real $y\}$. **[1 mark]**

You could write this as domain: $(-\infty, 0]$, range: $\{y: y \in \boldsymbol{R}\}$.

c This is a hyperbola in the 2nd and 4th quadrants with asymptotes at $x = -2$ and $y = 0$.



[2 marks]

Domain is $\{x: \text{all real } x \neq -2\}$; range is $\{y: \text{all real } y \neq 0\}$. **[1 mark]**

You could write this as domain: $\{x: x \in \boldsymbol{R}\backslash\{-2\}\}$, range: $\{y: y \in \boldsymbol{R}\backslash\{0\}\}$.

d This is a cubic function (polynomial of degree 3) with x-intercepts where:

$(x - 2)(x + 1)(x + 2) = 0$

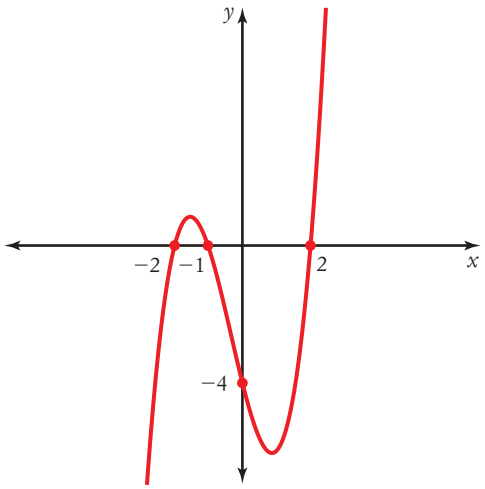
$x = 2, -1, -2$

For $x > 2$: $y > 0$

For $-1 < x < 2$: $y < 0$

For $-2 < x < -1$: $y > 0$

For $x < -2$: $y < 0$

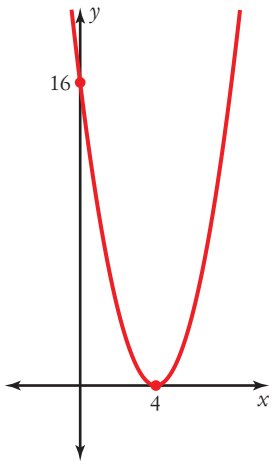


[3 marks]

Domain is $\{x: \text{all real } x\}$; range is $\{y: \text{all real } y\}$. **[1 mark]**

You could write this as domain: $\{x: x \in \boldsymbol{R}\}$, range: $\{y: y \in \boldsymbol{R}\}$.

e This is a parabola with x-intercept 4 and turning point (4, 0). It is a horizontal translation of $y = x^2$ by 4 units to the right.



[2 marks]

Domain is $\{x: \text{all real } x\}$; range is $\{y: \text{all real } y \geq 0\}$. **[1 mark]**

You could write this as domain: $\{x: x \in \boldsymbol{R}\}$, range: $(0, \infty]$.

2 a Substitute $m = 3$, $(4, 0)$ into $y - y_1 = m(x - x_1)$

$y - 0 = 3(x - 4)$ **[1 mark]**

$y = 3x - 12$ **[1 mark]**

or $3x - y - 12 = 0$

b Substitute $(-2, -1)$ into $y = a(x - b)^2 + c$

$y = a[x - (-2)]^2 + (-1)$

$y = a(x + 2)^2 - 1$ **[1 mark]**

Substitute $(0, 11)$ into $y = a(x + 2)^2 - 1$

$11 = a(0 + 2)^2 - 1$ **[1 mark]**

$12 = 4a$

$3 = a$

$y = 3(x + 2)^2 - 1$ **[1 mark]**

$= 3(x^2 + 4x + 4) - 1$

$= 3x^2 + 12x + 12 - 1$

$y = 3x^2 + 12x + 11$ **[1 mark]**

c Substitute $r = 2$ and $(-3, 2)$ into $(x - a)^2 + (y - b)^2 = r^2$

$[x - (-3)]^2 + (y - 2)^2 = 2^2$ **[1 mark]**

$(x + 3)^2 + (y - 2)^2 = 4$

$x^2 + 6x + 9 + y^2 - 4y + 4 = 4$ **[1 mark]**

$x^2 + 6x + y^2 - 4y + 13 = 4$

$x^2 + 6x + y^2 - 4y + 9 = 0$ **[1 mark]**

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3 a $f(-x) = (-x)^3 - 4(-x)$

$= -x^3 + 4x$

$= -(x^3 - 4x)$ **[1 mark]**

$= -f(x)$

Since $f(x) = -f(x)$. the function is odd. **[1 mark]**

b $f(-1) = (-1)^3 - 4(-1) = 3$ **[1 mark]**

c When $x = 0$:

$f(0) = (0)^3 - 4(0) = 0$

So y-intercept is 0. **[1 mark]**

When $y = 0$:

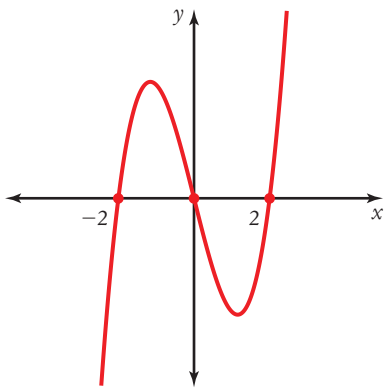
$0 = x^3 - 4x$

$= x(x^2 - 4)$

$x = 0, x^2 - 4 = 0$

$x = 0, x = \pm 2$ **[1 mark]**

d



[2 marks]

4 a $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

$= \left(\frac{-2 + (-1)}{2}, \frac{3 + (-4)}{2}\right)$ **[1 mark]**

$= \left(\frac{-3}{2}, \frac{-1}{2}\right)$

$= \left(-1\frac{1}{2}, -\frac{1}{2}\right)$ **[1 mark]**

b $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$= \sqrt{(-1 - (-2))^2 + (-4 - 3)^2}$ **[1 mark]**

$= \sqrt{1^2 + (-7)^2}$

$= \sqrt{1 + 49}$

$= \sqrt{50}$

$= \sqrt{25} \times \sqrt{2}$

$= 5\sqrt{2}$ units **[1 mark]**

c $m = \dfrac{y_2 - y_1}{x_2 - x_1}$

$= \dfrac{-4 - 3}{-1 - (-2)}$ **[1 mark]**

$= \dfrac{-7}{1}$

$= -7$ **[1 mark]**

5 a $\triangle = b^2 - 4ac$

$= (-4)^2 - 4(1)(-5)$ **[1 mark]**

$= 36$ **[1 mark]**

Since $\triangle > 0$ and is a perfect square, there are 2 rational zeros. **[1 mark]**

b $x = -\dfrac{b}{2a}$

$= -\dfrac{-4}{2 \times 1}$

$= 2$

The axis of symmetry has the equation $x = 2$. **[1 mark]**

c Substitute $x = 2$.

$y = 2^2 - 4(2) - 5$

$= -9$

The turning point is $(2, -9)$. **[1 mark]**

It is a minimum since $a > 0$ $(a = 1)$. **[1 mark]**

6 $x^2 + 4x + y^2 - 6y - 12 = 0$

$x^2 + 4x + y^2 - 6y = 12$

$x^2 + 4x + 4 + y^2 - 6y + 9 = 12 + 4 + 9$ **[1 mark]**

$(x + 2)^2 + (y - 3)^2 = 25$ **[1 mark]**

$[x - (-2)]^2 + (y - 3)^2 = 5^2$

This is in the form $(x - a)^2 + (y - b)^2 = r^2$

Radius is 5 units. **[1 mark]**

Centre is $(-2, 3)$. **[1 mark]**

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7 $y = 3x - 1$ has $m_1 = 3$

$6x - 2y - 5 = 0$

$6x - 5 = 2y$

$3x - 2.5 = y$

$m_2 = 3$ **[1 mark]**

Since $m_1 = m_2$, the lines are parallel. **[1 mark]**

Part C

1 a $y = ax^2 + bx + c$ has x-intercepts at $y = 0$.

For $ax^2 + bx + c = 0$, $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ **[1 mark]**

b The axis of symmetry is halfway between the x-intercepts x_1 and x_2, i.e., $\dfrac{x_1 + x_2}{2}$

$$x = \frac{\dfrac{-b - \sqrt{b^2 - 4ac}}{2a} + \dfrac{-b + \sqrt{b^2 - 4ac}}{2a}}{2}$$ **[1 mark]**

$$= \frac{\dfrac{-2b}{2a}}{2}$$ **[1 mark]**

$$= \frac{\dfrac{-b}{a}}{2}$$

$$= -\frac{b}{2a}$$ **[1 mark]**

2 a $3x - 4y + 5 = 0$ [1]

$x^2 + y^2 = 25$ [2]

From [1]:

$3x + 5 = 4y$

$\dfrac{3x + 5}{4} = y$ [3] **[1 mark]**

Substitute [3] into [2]:

$$x^2 + \left(\frac{3x + 5}{4}\right)^2 = 25$$ **[1 mark]**

$$x^2 + \frac{9x^2 + 30x + 25}{16} = 25$$

$16x^2 + 9x^2 + 30x + 25 = 400$

$25x^2 + 30x - 375 = 0$ **[1 mark]**

$5x^2 + 6x - 75 = 0$

$\triangle = b^2 - 4ac$

$= 6^2 - 4(5)(-75)$

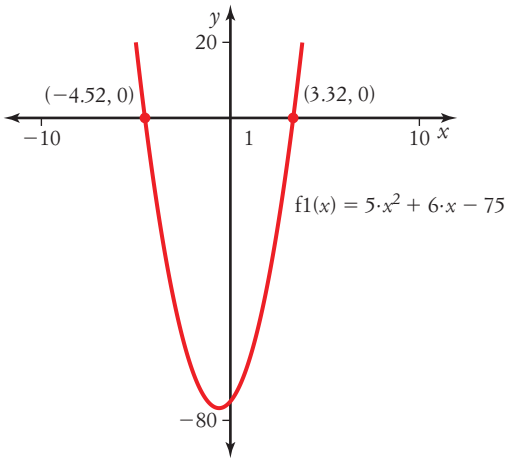
$= 1536$

> 0 (2 zeros)

The line crosses the circle at 2 points. **[1 mark]**

b On the CAS calculator sketch $y = 5x^2 + 6x - 75$ and find the x-intercepts where $y = 0$.

Namely, $5x^2 + 6x - 75 = 0$ **[1 mark]**



$x = -4.5, 3.3$ **[2 marks]**

Substitute $x = -4.5$ into $3x - 4y + 5 = 0$

$3(-4.5) - 4y + 5 = 0$ **[1 mark]**

$-4y - 8.5 = 0$

$-8.5 = 4y$

$-2.1 = y$ **[1 mark]**

Substitute $x = 3.3$ into $3x - 4y + 5 = 0$.

$3(3.3) - 4y + 5 = 0$ **[1 mark]**

$-4y + 14.9 = 0$

$14.9 = 4y$

$3.7 = y$ **[1 mark]**

The points of intersection are $(-4.5, -2.1)$ and $(3.3, 3.7)$. **[1 mark]**

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